

SHREYA ASHAR

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Education

Sardar Vallabhbhai National Institute of Technology, Surat

Expected Apr 2028

Bachelor of Technology in Artificial Intelligence

CGPA: 8.35/10

- Relevant Coursework: **Web Programming, Python, Data Structures, Object-Oriented Programming, Artificial Intelligence, Theory of Computation, Operating Systems, DBMS, Computer Networks**

Projects

AirHelp | *GitHub*

Technologies: Python, FastAPI, React, ChromaDB, NetworkX, SQLite

- Developed a full-stack airport assistance platform integrating **Retrieval-Augmented Generation (RAG)**, **Vector Search**, and **Information Retrieval** for passenger support.
- Designed a **Graph-Based Navigation System** using **Shortest Path Algorithms** on 109 nodes and 120 weighted edges for indoor route optimization.
- Built multimodal pipelines for **OCR**, **Speech Processing**, and **Real-Time Notifications**, enabling efficient event-driven workflows.

SemEval-2026: Narrative Story Similarity | *GitHub*

Technologies: PyTorch, Hugging Face, SBERT, DeBERTa, Contrastive Learning

- Developed a semantic similarity system using hybrid **SBERT Embeddings**, **TF-IDF**, and **Transformer-based Models** for narrative comparison.
- Built a **Siamese Neural Network** with **Contrastive Learning** for pairwise ranking on 2000+ samples.
- Fine-tuned a **DeBERTa Cross-Encoder** using regression and contrastive objectives for continuous similarity prediction.
- Optimized **Model Training** using mixed precision, gradient accumulation, and systematic hyperparameter tuning.

MindSettler | *Website* | *GitHub*

Technologies: Next.js, React, TypeScript, Django, Firebase, Docker, SQLite

- Built a full-stack **Mental Wellness Platform** using **Next.js** and **Django** with scalable **REST APIs**.
- Implemented secure **Authentication**, **Authorization**, email verification, and session management using Django Auth.
- Integrated a real-time chatbot using **Firebase Functions** and containerized chatbot services with **Docker**.

LogiFlow: Google Solution Challenge | *Website* | *GitHub*

Technologies: Python, FastAPI, Next.js, React, CatBoost, Scikit-Learn, PostgreSQL

- Developed an **AI-powered multimodal logistics optimization platform** that generates optimal **road, rail, water, air, and hybrid transport** routes using multi-objective optimization and constraint-based planning.
- Built **Machine Learning** pipelines using **CatBoost** with feature engineering to predict shipment delays by integrating weather, railway operations, and transport-specific data.
- Designed modular **FastAPI** microservices implementing **graph algorithms**, Pareto optimization, risk scoring, dynamic pricing, and real-time route recommendations for intelligent logistics planning.

Technical Skills

Programming Languages: Python, C++, Java, JavaScript, C

Machine Learning: PyTorch, Hugging Face Transformers, SBERT, DeBERTa, Contrastive Learning, NLP, Information Retrieval

Technologies/Frameworks: React, Next.js, Node.js, Express.js, Django, Flask, FastAPI, Tailwind CSS

Databases: MongoDB, MySQL, SQLite

Skills: Data Structures & Algorithms, Object-Oriented Programming, REST APIs, Git, GitHub, Problem Solving

Achievements

- **Selected Among Top 106 Teams Worldwide** — Google Solution Challenge 2026 (from 6,700+ teams)
- **Winner** — Web Wonders 3.0 2025
- **Finalist** — IIIT Lucknow Hackathon 2025
- **Contributor** — GirlScript Summer of Code 2025

Positions of Responsibility

Junior Developer, Google Developer Groups on Campus, SVNIT

Nov 2025 – Present

Executive, ACM Student Chapter, SVNIT

Nov 2025 – Present

Member, Nexus Tech Cell, SVNIT

Sep 2024 – Present